Game Score Prediction and Critics Analysis: ML-Based Analysis on Video Game Ratings

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***Abstract*—Introduction/Background: Media review scores play a significant role in shaping a game's commercial success and public perception. However, it remains unclear whether these scores reflect game quality alone or are also shaped by content features such as female representation and character sexualization.**

Purpose: This project examines whether media review scores for video games are influenced by features such as female representation, sexualization, genre, country, and platform.

Methods: We build predictive models including Lasso Regression and Ridge Regression for continuous score prediction, and Naive Bayes for binary classification, to estimate ratings and identify potential biases. Performance is evaluated using MSE, RMSE, R^2, accuracy, precision, recall, and F1-score.

Expected Results: We expect to identify key predictors of ratings and detect potential bias linked to sexualization or representation features.

Significance: This study may highlight how media evaluations reflect underlying preferences, contributing to discussions on fairness and representation in the gaming industry, or potentially providing empirical evidence against such critiques.

I. Introduction

Media review scores strongly influence a video game's visibility and commercial success. However, the mechanism behind mainstream media scores remains unclear. Features such as female representation and sexualization may significantly shape reviewers' perceptions, and understanding whether such biases exist is important for promoting fairness and transparency in game development, especially for future generations of gamers.

Most prior work focuses on descriptive statistics of gender representation, leaving a gap in statistical modeling. There is also limited use of machine learning to systematically evaluate whether content-related features influence ratings across multiple attributes such as genre, platform, and issuing country.

This project proposes using Lasso Regression and Ridge Regression to predict continuous review scores, alongside Naive Bayes for classifying high versus low ratings. These methods are chosen for their interpretability and ability to handle feature relationships. Model performance will be evaluated using MSE, RMSE, R^2, accuracy, precision, recall, and F1-score.

We will also develop a Streamlit application that allows users to upload a dataset, select model configurations, and receive predicted scores alongside insights into which factors most influence ratings. The app targets gamers, developers, and researchers interested in understanding review score patterns and potential biases. By applying predictive analytics to questions of representation, this research can help identify systematic preferences in game evaluations and inform more equitable review practices.

Broader Impact: This work contributes a quantitative, ML-driven framework for auditing critic review scores. The methodology can generalize beyond video games to other media such as film and music where similar representation concerns have been raised.

II. Background

A. Comparisons to Prior Work

Prior work on video game ratings and machine learning falls into two streams: predicting scores from game metadata and analyzing gender representation. On the prediction side, early work by Greenwood-Ericksen et al. questioned whether Metacritic scores validly reflect game quality [1]. Subsequent ML studies demonstrated that features such as genre, platform, and user review volume can predict critic scores with meaningful accuracy. Studies applying Ridge, Lasso, and linear regression models to games datasets achieved R^2 scores in the range of 0.68–0.83, with nonlinear models outperforming linear ones, suggesting that the relationship between game features and ratings is not purely additive [2]. More recently, research integrating Steam, Twitch, and Metacritic data confirmed that critic scores exhibit a significant positive correlation with game revenue, though their predictive weight is lower than community-engagement signals such as follower count and peak concurrent users [3].

On the representation side, Rennick et al. documented that female characters speak approximately half as much as male characters and are more likely to be sexualized and assigned secondary roles [4]. De la Torre-Sierra and Guichot-Reina further confirm that women are often portrayed with unrealistic body proportions or subjected to various forms of violence [5]. A systematic literature review of gender bias in video game character design identified recurring patterns of hypersexualized models and biased gameplay mechanics across MOBA and console titles, while noting that few studies propose practical frameworks to mitigate this bias [6]. A study on gender stereotypes in Overwatch applied an Ensemble Late Fusion method across multiple classifiers to predict whether player behaviors reflected gender-based stereotypes, finding that many players exhibited biased attitudes toward female players [7].

B. Common Knowledge and Disputes

A key unresolved question in the field is whether media review scores reflect objective gameplay quality or are also shaped by content-related attributes such as representation. Research on the "review bombing" phenomenon, in which users assign unusually low scores to critically acclaimed games, found a threshold gap of 4 or more points between professional Metascores and average user ratings across 68 PC titles. This suggests that professional and user evaluations are systematically driven by different factors [8], raising the broader question of what critics actually reward.

The ML fairness literature has also established that biased training data can propagate and amplify existing social inequities. A systematic review of 120 papers on gender bias in ML found that gender-unbalanced datasets do not necessarily introduce scoring bias but can enlarge gender disparities and reduce fairness in model outputs [9]. Whether these dynamics translate to game review datasets, where the data reflects human critic judgment rather than automated labels, remains an open empirical question this project directly addresses.

C. Knowledge Gaps

Existing studies on video game ratings use structural features like genre and sales figures but do not incorporate character-level representation data. Conversely, representation studies rely on descriptive statistics rather than predictive modeling. This project bridges that gap by engineering game-level features from character-level data (female character ratio, sexualization proportion) and combining them with structural attributes in regularized regression and Naive Bayes models. Unlike prior rating-prediction work that treats review scores as purely quality signals, we treat them as potentially biased outputs and make bias analysis a primary goal. Our dataset of 64 games is smaller than those used in prior prediction studies, which motivates our use of regularized models (Ridge, Lasso) that are known to generalize well under limited samples. This tradeoff between sample size and model complexity is not directly addressed in larger-data prior work.

III. End-to-End ML Pipeline

A. Offline Model Training and Evaluation

We will develop and test machine learning models primarily in Jupyter Notebooks. In these notebooks, we will perform data cleaning, feature creation, preprocessing, model training, and evaluation experiments. Different models and configurations will be compared offline, and the best-performing model will be deployed in the Streamlit application.

1) Data Collection, Exploration, and Processing: We use the *Gender Representation in Video Games* dataset from Kaggle, released in November 2022 by Brisa Palomar. It includes 64 games, 637 characters, and 637 corresponding sexualization records across three linked tables. The dataset contains both numerical features (review scores, team size, number of protagonists) and categorical features (genre, platform, country, director gender, character gender). The main target variable is the average critic review score (`Avg_reviews`).

We will aggregate character-level and sexualization-level data into game-level features such as female character ratio, non-male character ratio, average sexualization score, and proportion of sexualized characters. These features will be merged with game-level attributes including genre, platform, PEGI rating, and country. Ground-truth labels are provided in the dataset. For regression, the label is `Avg_reviews`. For classification, scores will be binarized at the median into high and low categories.

Exploratory analysis will include histograms of review score distributions, bar charts of genre and platform frequencies, scatterplots of female character ratio versus review score, and boxplots comparing sexualization scores across rating categories. The Streamlit application will allow users to interactively filter data and view updated visualizations.

Preprocessing steps include merging the three tables, handling missing values via median imputation, one-hot encoding of categorical variables (genre, platform, country), normalization of numerical features to zero mean and unit variance, correlation analysis to identify redundant features, and outlier removal using IQR-based thresholds. These steps ensure the dataset is suitable for machine learning and help improve model generalization.

2) Methods and Model Training: We implement Ridge Regression, Lasso Regression, and Naive Bayes using NumPy and Pandas only, without any high-level ML libraries.

Ridge Regression minimizes the residual sum of squares with an ℓ_2 penalty to control overfitting:

$$\hat{\beta}_{\text{Ridge}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2 + \lambda \|\beta\|_2^2 \tag{1}$$

The closed-form solution is:

$$\hat{\beta} = (X^\top X + \lambda I)^{-1} X^\top y \tag{2}$$

Lasso Regression applies an ℓ_1 penalty, inducing sparsity and enabling automatic feature selection:

$$\hat{\beta}_{\text{Lasso}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2 + \lambda \|\beta\|_1 \tag{3}$$

Naive Bayes uses Bayes' theorem with a conditional independence assumption to classify games as high or low rated:

$$P(y \mid x) \propto P(y) \prod_{j=1}^{d} P(x_j \mid y) \tag{4}$$

These models are well-suited because they handle multicollinearity, provide interpretable coefficients, and generalize effectively on small datasets. Lasso also performs implicit feature selection, which is useful for identifying the most influential representation variables.

Model inputs are game-level feature vectors $x \in \mathbb{R}^d$ comprising numerical and one-hot encoded categorical features. For regression, the output is a continuous predicted review score $\hat{y} \in \mathbb{R}$. For Naive Bayes, the output is a binary class label $\hat{y} \in \{0, 1\}$.

3) Model Evaluation: **Regression metrics:**

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{5}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{6}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{7}$$

Classification metrics:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{8}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

MSE and RMSE penalize large prediction errors in review scores, R^2 measures how much variance our model explains, and F1-score balances precision and recall for the binary classification task. Together these metrics give a complete picture of model performance across both problem formulations.

We will split the dataset 70/15/15 into training, validation, and test sets. The regularization hyperparameter λ will be tuned on the validation set using a grid search over $\{0.01, 0.1, 1, 10, 100\}$. Regularization and cross-validation will be used to avoid overfitting, while multiple λ configurations will be tested to prevent underfitting.

4) Model Deployment: The final model will be selected based on a balance between predictive performance and interpretability. If Ridge and Lasso achieve similar R^2 scores, Lasso will be preferred for its feature selection properties, which make the bias analysis easier to interpret. The selected model will be serialized and integrated into the Streamlit back-end.

B. Front-End Application Using Streamlit

Our web application targets gamers, developers, and researchers interested in understanding review score patterns and potential biases.

The UI is organized into four pages navigable via a sidebar: *Home*, *EDA*, *Model Training*, and *Predict & Analyze*. The sidebar also contains persistent configuration controls including a CSV file uploader, a model selection dropdown (Ridge, Lasso, Naive Bayes), a feature selection checklist, and a regularization slider for λ. A *Run Prediction* button triggers the full pipeline.

The *EDA* page displays dataset previews, histograms, scatterplots, and boxplots generated from the uploaded data. The *Model Training* page shows training progress and validation curves. The *Predict & Analyze* page displays four metric cards (R^2, MSE, RMSE, F1), a feature importance bar chart, a predicted vs. actual scatter plot, a bias analysis table showing signed coefficients for representation features, and an interactive prediction panel where users can adjust sliders for female character ratio and sexualization score to see live predicted score updates.

Figure 1 shows the proposed layout.

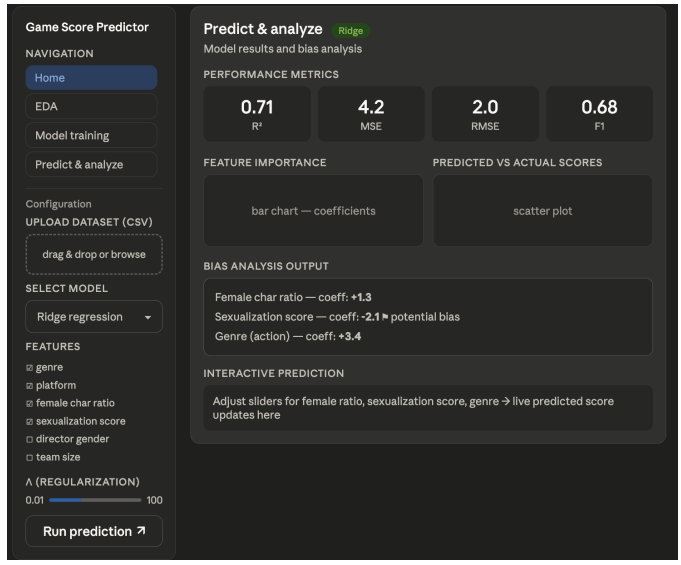

Fig. 1. Proposed Streamlit UI layout. The sidebar (left) contains dataset upload, model selection, feature checkboxes, λ slider, and the Run Prediction button. The main panel (right) shows performance metric cards, a feature importance chart, a predicted vs. actual scatter plot, bias analysis output, and an interactive live prediction panel.

When the user uploads a dataset and clicks Run Prediction, the back-end applies the same preprocessing transformations used during offline training (missing value imputation, one-hot encoding, normalization), executes the selected model, and returns evaluation metrics and coefficient values to the front-end for display. The full system pipeline flows from the user interface through data preprocessing and model training to evaluation and results display.

IV. Risk & Mitigation

A key challenge is the small dataset size (64 games), which increases the risk of overfitting. We will mitigate this using L1/L2 regularization, a held-out test set, and cross-validation. Feature engineering complexity, particularly aggregating character-level data to the game level, will be addressed through careful merge validation and exploratory sanity checks. Multicollinearity among representation features will be handled through Ridge regression. We also acknowledge that results represent correlation rather than causation and will interpret findings cautiously throughout.

V. Expected Outcomes

We expect Ridge and Lasso regression to achieve R^2 scores above 0.6, consistent with prior rating prediction work using small structured datasets [2]. We also expect Lasso to surface a sparse set of influential features, with genre and platform emerging as primary predictors based on prior work [3], while representation features (female ratio, sexualization score) contribute secondary but statistically detectable effects. Finally, the Naive Bayes classifier is expected to achieve F1-scores in the range of 0.65–0.75 for binary rating classification, pro-

viding a complementary view of which feature combinations separate high- from low-rated games.

VI. Team Member Contributions

A. Technical Components

- **Maggie:** Data preprocessing pipeline — table merging, missing value handling, one-hot encoding, normalization, and feature engineering from character-level data.
- **Vivian:** Offline model training — NumPy/Pandas implementation of Ridge Regression, Lasso Regression, and Naive Bayes; hyperparameter tuning.
- **Rui:** Model evaluation — implementation of MSE, RMSE, R^2, accuracy, precision, recall, F1-score; validation procedures and experiment logging.
- **Yujun:** Front-end development — Streamlit application layout, sidebar controls, EDA visualizations, and results display pages.
- **Xueer:** Risk analysis and back-end integration — connecting the preprocessing and model pipeline to the Streamlit front-end.
- **Samy:** System integration, LaTeX formatting, and cross-component testing.

B. Writing Components

- **Samy:** LaTeX compilation, proofreading, abstract, and overall document formatting.
- **Yujun:** Introduction and Streamlit application description.
- **Xueer:** Risk & Mitigation and Expected Outcomes sections.
- **Maggie:** Data Collection, Exploration & Processing subsection.
- **Vivian:** Methods and Model Training subsection.
- **Rui:** Model Evaluation subsection.

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